LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 7: Variables and Operators

Module 3 | 45 minutes | Variables, input and operators | Grades 6–8 · ages 11–14

Objective: Store and update data in variables, read user input, and use arithmetic and join operators to calculate and build messages.

Success criteria

- I can store and update data in variables.
- I can use user input.
- I can calculate with operators and build messages with join.

Key vocabulary

- **variable**: A named place to store a value.
- **assign**: Give a variable a value (set).
- **input**: Data the user gives the program.
- **operator**: A symbol that calculates or combines values.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Jar trace (5 min): labelled jars and counters; follow set/change instructions read aloud.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Counter and calculator: learners follow the steps.
14 min	**Perform**	Code Lab: click counter, Code Lab: adding machine, Trace the variable
9 min	**Probe**	Variables in Python, activity
4 min	**Wrap-up**	Exit ticket: What value is in score after: set score to 3, change score by 4?

Unplugged warm-up

Jar trace (5 min): labelled jars and counters; follow set/change instructions read aloud.

Teaching the idea

Variables are named storage; set assigns, change increments. Input via ask/answer must be stored before the next ask. Arithmetic and string (join) operators compute outputs. Python bridge shows assignment and input.

Analogy: Labelled jars: the label is the name, what's inside is the value — you can swap it any time.

Live demo: Trace a variable on the board while running a program step by step.

Questions to ask

Ask	Listen for
What does change do?	Adds to the current value.
Why copy answer into a variable?	The next ask overwrites answer.
What does join do?	Sticks text together.

Differentiation

Support
Provide the asks; learners add the variables and operator.

Stretch
Make a calculator that also says the difference, product and quotient.

Common misconceptions

Misconception	How to fix it
A variable keeps every value it had	It holds one value at a time.
y = x * 3 updates when x changes	It is calculated once, when it runs.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What value is in score after: set score to 3, change score by 4?

Homework
Write a plan for a "pocket money calculator": what inputs, variables and operators would it need?

Answer key — Kitabash activities

1. Code Lab: click counter (perform · code lab)

Goals checked: Counts 3 clicks · Starts at 0 · Shows the count

One solution:

Bot

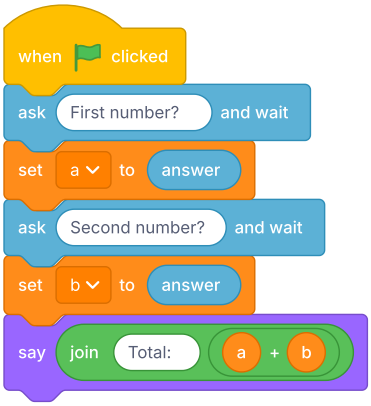


2. Code Lab: adding machine (perform · code lab)

Goals checked: 7 + 5 = 12 · 20 + 22 = 42 · Builds a message

One solution:

Bot



3. Trace the variable (perform · multiple choice)

Answer: 5

4. Variables in Python (probe · code output)

5. Explain your thinking (probe · free response)

Look for: set, change, value. **Example:** Set replaces the value, so score becomes exactly 5. Change adds to the value, so score goes up by 5. I would use set at the start of a game to reset the score, and change when the player earns points.

Answer key — worksheet

1. **Key words:** variable = A named place to store a value. · assign = Give a variable a value (set). · input = Data the user gives the program. · operator = A symbol that calculates or combines values.
2. **Plan your code:** Goals: Counts 3 clicks; Starts at 0; Shows the count.
3. **Plan your code:** Goals: 7 + 5 = 12; 20 + 22 = 42; Builds a message.
4. **Trace the variable:** A. 5 — Set replaces the value (2), then change adds 3 → 5.
5. **Variables in Python:** A. 5 12

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